

Phase 7: Complete Thermistor Monitoring System

Objective

Integrate thermistor temperature sensing with the complete CPU system to create a real-time temperature monitoring application with digital display and analog output.

Component BOM

Thermistor Interface

Qty	Component	Function
1	NTC Thermistor (10kΩ @ 25°C)	Temperature sensor
1	10kΩ precision resistor	Voltage divider reference
1	LM358 op-amp	Signal conditioning
2	10kΩ resistors	Op-amp biasing
1	100kΩ resistor	Op-amp feedback
2	0.1μF capacitors	Op-amp decoupling

Analog-to-Digital Conversion

Qty	Component	Function
1	ADC0808	8-bit ADC with multiplexer
1	74F04	Clock generation for ADC
1	10μF capacitor	ADC reference filtering
3	0.1μF capacitors	ADC decoupling

System Integration

Qty	Component	Function
1	74LS244	ADC data buffer
1	74F08	Control logic gates
1	Precision voltage reference (2.5V)	ADC reference
-	Additional breadboard space	Sensor integration

System Architecture

Temperature Sensing Chain

Thermistor → Voltage Divider → Op-Amp Buffer → ADC → CPU

↓ ↓ ↓ ↓ ↓

10kΩ NTC Reference Signal 8-bit Processing
Sensor Resistor Conditioning Data & Display

Memory Map Extensions

0xA100: ADC control register

0xA101: ADC data register

0xA102: Temperature sensor configuration

0xA103: Calibration offset register

0xA104: Calibration scale register

Temperature Calculation

Raw ADC Value → Voltage → Resistance → Temperature

Voltage = $(\text{ADC_Value} / 255) \times \text{Reference_Voltage}$

Resistance = $R_{\text{ref}} \times (V_{\text{ref}} - V_{\text{sensor}}) / V_{\text{sensor}}$

Temperature = Steinhart-Hart equation or lookup table

Circuit Implementation

Thermistor Signal Conditioning

Thermistor Voltage Divider:

+5V → 10kΩ Reference Resistor → Measurement Point → Thermistor → GND

Op-Amp Buffer (LM358):

Non-inverting input ← Thermistor voltage

Inverting input ← Output (unity gain buffer)

Output → ADC input channel 0

Temperature Range: 0°C to 100°C

Voltage Range: ~1V to 4V (depending on thermistor curve)

ADC Interface

ADC0808 Configuration:

Channel 0: Thermistor input

Channels 1-7: Reserved for future sensors

Reference: 2.5V precision reference

Resolution: $2.5V / 256 = \sim 9.8mV$ per LSB

Control Interface:

Address lines select input channel

Start conversion pulse initiates reading

End of conversion signals data ready

CPU Integration

ADC Control:

CPU writes to 0xA100 → Start conversion

CPU polls 0xA100 → Check conversion complete

CPU reads 0xA101 → Get temperature data

Data Processing:

Raw ADC → Lookup table → Temperature value

Temperature → 7-segment display

Temperature → DAC output (0-4V for 0-100°C)

Enhanced Software Architecture

Temperature Measurement Routine

asm

; Read temperature from thermistor

READ_TEMPERATURE:

; Start ADC conversion

LD R0,#0x01 ; Start conversion command

OUT 0xA100,R0 ; Send to ADC control

; Wait for conversion complete

WAIT_ADC:

IN R0,0xA100 ; Read ADC status

AND R0,#0x80 ; Check EOC bit

JZ WAIT_ADC ; Loop until complete

; Read temperature data

IN R0,0xA101 ; Read ADC result

ST R0,TEMP_RAW ; Store raw reading

; Convert to temperature

CALL ADC_TO_TEMP ; Convert using lookup table

ST R0,TEMP_CELSIUS ; Store final temperature

RET

Temperature Conversion Algorithm

asm

```
; Convert ADC reading to Celsius temperature
; Input: R0 = ADC value (0-255)
; Output: R0 = Temperature in Celsius
```

ADC_TO_TEMP:

```
; Use lookup table for thermistor linearization
LD R1,#TEMP_TABLE_BASE ; Load table address
ADD R1,R0                ; Index into table
LD R0,[R1]               ; Get temperature value
```

```
; Apply calibration offset
LD R1,CALIBRATION_OFFSET
ADD R0,R1                ; Adjust for system offset
```

```
; Apply calibration scale
LD R1,CALIBRATION_SCALE
MUL R0,R1                ; Scale correction
DIV R0,#100              ; Normalize scaling
```

RET

Display and Output Management

asm

```
; Main temperature monitoring loop
```

TEMP_MONITOR:

```
CALL READ_TEMPERATURE ; Get current temperature
```

```
; Display on 7-segment
LD R0,TEMP_CELSIUS    ; Load temperature value
CALL DISPLAY_TEMP      ; Show on display
```

```
; Output to DAC
LD R0,TEMP_CELSIUS    ; Load temperature
CALL TEMP_TO_DAC       ; Convert and output
```

```
; Delay before next reading
CALL DELAY_1SEC        ; 1 second delay
```

```
JMP TEMP_MONITOR      ; Repeat forever
```

Calibration Routine

asm

; Two-point calibration using known temperatures

CALIBRATE_SYSTEM:

; Calibration point 1: Ice water (0°C)

CALL PROMPT_ICE_WATER ; User instruction

CALL READ_RAW_ADC ; Get ADC reading

ST R0,CAL_POINT_0 ; Store 0°C reading

; Calibration point 2: Boiling water (100°C)

CALL PROMPT_BOILING ; User instruction

CALL READ_RAW_ADC ; Get ADC reading

ST R0,CAL_POINT_100 ; Store 100°C reading

; Calculate calibration parameters

CALL CALC_CAL_PARAMS ; Determine offset/scale

RET

Build Steps

1. Thermistor Interface Construction

- Build voltage divider with thermistor and reference resistor
- Add LM358 op-amp buffer circuit
- Test voltage output vs. temperature manually
- Verify linear response range

2. ADC Integration

- Install ADC0808 with precision voltage reference
- Connect thermistor output to ADC channel 0
- Add control interface to CPU I/O system
- Test ADC conversion timing and accuracy

3. Software Development

- Implement ADC control routines
- Create temperature conversion lookup table

- Develop display formatting functions
- Add DAC output scaling

4. System Calibration

- Implement calibration routine using known temperatures
- Create calibration data storage in EEPROM
- Verify accuracy across temperature range
- Document calibration procedure

5. Application Integration

- Combine all subsystems into complete application
- Add error handling and bounds checking
- Implement averaging for noise reduction
- Test complete system operation

Test Procedures

1. Thermistor Response Test

Objective: Verify thermistor voltage vs. temperature

Equipment: Thermometer, ice water, warm water, multimeter

Procedure:

1. Measure voltage at room temperature ($\sim 25^{\circ}\text{C}$)
2. Place thermistor in ice water, measure voltage
3. Place thermistor in warm water (50°C), measure voltage
4. Plot voltage vs. temperature curve

Expected Results:

- Ice water: $\sim 3.8\text{V}$
- Room temp: $\sim 2.5\text{V}$
- Warm water: $\sim 1.2\text{V}$

2. ADC Accuracy Test

Objective: Verify ADC conversion accuracy

Test Setup: Precision voltage source, multimeter

Procedure:

1. Apply known voltages to ADC input (0V, 1.25V, 2.5V)
2. Read ADC output via CPU
3. Calculate conversion accuracy
4. Test all 8 ADC channels

Expected Results: ± 1 LSB accuracy ($\pm 9.8\text{mV}$)

3. Temperature Measurement Test

Objective: Verify end-to-end temperature accuracy

Test Program:

```
; Continuous temperature readout
TEMP_TEST:
    CALL READ_TEMPERATURE
    CALL DISPLAY_TEMP    ; Show on 7-segment
    CALL SERIAL_OUTPUT   ; Send to serial port (if available)
    CALL DELAY_500MS     ; Update rate: 2Hz
    JMP TEMP_TEST
```

Validation: Compare CPU readings with calibrated thermometer

4. System Response Time Test

Objective: Measure temperature measurement latency

Procedure:

1. Apply sudden temperature change (ice $\rightarrow$ room temp)
2. Monitor time to 90% final reading
3. Measure total system delay
4. Verify acceptable response time

Expected Results: < 5 second response time

5. Accuracy and Stability Test

Objective: Long-term measurement validation

Test Duration: 24 hours continuous operation

Measurements:

- Temperature drift over time
- Measurement noise and stability
- System reliability and error rate
- Power consumption

Application Programs

Basic Temperature Monitor

```
asm
```

```
; Simple temperature display application
```

```
MAIN:
```

```
    CALL SYSTEM_INIT    ; Initialize hardware
```

```
MAIN_LOOP:
```

```
    CALL READ_TEMPERATURE ; Get current temp
```

```
    CALL DISPLAY_TEMP     ; Show on display
```

```
    CALL OUTPUT_DAC       ; Analog output
```

```
    CALL DELAY_1SEC       ; Update rate
```

```
    JMP MAIN_LOOP
```

Temperature Logger

asm

; Store temperature readings in memory

LOGGER:

LD R3,#LOG_START ; Initialize log pointer

LOG_LOOP:

CALL READ_TEMPERATURE ; Get reading

ST R0,[R3] ; Store in memory

INC R3 ; Advance pointer

; Check for memory full

LD R1,#LOG_END

CMP R3,R1

JZ LOG_FULL

CALL DELAY_10SEC ; Log interval

JMP LOG_LOOP

LOG_FULL:

CALL DUMP_LOG ; Output stored data

JMP LOGGER ; Restart logging

Temperature Alarm System

asm

; Monitor for temperature thresholds

ALARM_MONITOR:

CALL READ_TEMPERATURE

; Check high temperature alarm

LD R1,HIGH_TEMP_LIMIT

CMP R0,R1

JGE HIGH_ALARM

; Check low temperature alarm

LD R1,LOW_TEMP_LIMIT

CMP R0,R1

JLE LOW_ALARM

; Normal operation

CALL NORMAL_DISPLAY

JMP ALARM_MONITOR

HIGH_ALARM:

CALL FLASH_DISPLAY ; Visual alarm

CALL SOUND_BUZZER ; Audio alarm

JMP ALARM_MONITOR

LOW_ALARM:

CALL SLOW_FLASH ; Different alarm pattern

JMP ALARM_MONITOR

Success Criteria

- ☐ Thermistor produces linear voltage response to temperature
- ☐ ADC converts temperature voltage accurately ($\pm 1^\circ\text{C}$)
- ☐ Temperature display shows correct values
- ☐ DAC output scales proportionally with temperature
- ☐ System responds to temperature changes within 5 seconds
- ☐ Calibration procedure produces accurate results
- ☐ 24-hour operation shows stable, repeatable measurements
- ☐ Complete application runs without errors

Performance Specifications

Measurement Accuracy

- **Resolution:** 0.4°C (8-bit ADC with 100°C range)
- **Accuracy:** $\pm 1^\circ\text{C}$ after calibration
- **Repeatability:** $\pm 0.2^\circ\text{C}$
- **Temperature Range:** 0°C to 100°C

System Performance

- **Update Rate:** 1-10Hz (user configurable)
- **Response Time:** <5 seconds to 90% final value
- **Power Consumption:** <500mA @ 5V
- **Operating Duration:** Continuous (limited by component reliability)

Integration Notes for Future Enhancement

- Plan for multiple sensor inputs
- Consider data logging to external storage
- Design for remote monitoring capability
- Add network interface for IoT applications
- Implement advanced signal processing algorithms

Documentation Deliverables

- Complete temperature monitoring application code
- Calibration procedure documentation
- Performance test results and accuracy validation
- User operation manual
- Troubleshooting guide
- Final system integration schematic
- Bill of materials with supplier information
- Project completion summary and lessons learned